

Developing an Algorithm

What is an Algorithm?

An **algorithm** is a **finite sequence of steps** to solve a problem.

It must be:

- **Input/output:** Each algorithm must take zero, one or more quantities as input data produce one or more output values
- **Finiteness:** An algorithm must terminate in a finite number of steps
- **Definiteness:** Each step of the algorithm must be precisely and unambiguously stated
- **Correctness:** Output should be correct.
- **Generality:** Solve problems of a specific type for any input data

Steps in Developing an Algorithm

1. Understand the problem
2. Identify Input and Output
3. Design logic (steps)
4. Write the algorithm
5. Test with sample data

Even or Odd

Problem

Check whether a number is even or odd.

Input

n

Output

Even / Odd

Algorithm

1. Start
2. Read n
3. If $n \bmod 2 = 0$, then go to step 4, else go to step 5
4. Print "Even"
5. Print "Odd"
6. Stop

Sum of First n Natural Numbers

Problem

Find sum of first n numbers

Input

n

Output

Sum

Algorithm

1. Start
2. Read n
3. $\text{sum} \leftarrow 0$
4. Set $i \leftarrow 1$
5. If $i \leq n$, go to step 6, else go to step 8
6. $\text{sum} \leftarrow \text{sum} + i$
7. $i \leftarrow i + 1$, go to step 5
8. Print sum
9. Stop

Factorial of a Number

Problem

Find factorial of a number

Input

n

Output

$n!$

Algorithm

1. Start
2. Read n
3. $\text{fact} \leftarrow 1$
4. Set $i \leftarrow 1$
5. If $i \leq n$, go to step 6, else go to step 8
6. $\text{fact} \leftarrow \text{fact} \times i$
7. $i \leftarrow i + 1$, go to step 5
8. Print fact
9. Stop

Write an algorithm for the following

1. Write an algorithm to calculate the simple interest using the formula.

Simple interest = $P \times N \times R / 100$.

2. Write an algorithm to find the area of the triangle.